

$$27 \int \frac{2x^3 - 5x^2 + 46x + 98}{(x^2 + x - 12)^2} dx$$

$$28 \int \frac{-2x^4 - 3x^3 - 3x^2 + 3x + 1}{x^2(x+1)^3} dx$$

c Exer. 29–32: Using a computer algebra system, determine the partial fraction decomposition of the integrand and then evaluate the integral.

$$29 \int \frac{282x^3 + 1021x^2 - 509x - 398}{36x^4 + 96x^3 - 131x^2 - 71x + 70} dx$$

$$30 \int \frac{-1302x^3 + 4075x^2 + 1742x - 13}{1980x^4 - 1641x^3 - 2684x^2 - 849x - 70} dx$$

$$31 \int \frac{-329x^2 - 440x + 4570}{30x^3 + 187x^2 - 555x - 250} dx$$

$$32 \int \frac{3244x^2 + 437x - 57}{6188x^3 + 1574x^2 - 420x + 18} dx$$

Exer. 33–36: Use partial fractions to evaluate the integral (see Formulas 19, 49, 50, and 52 of the table of integrals in Appendix II).

$$33 \int \frac{1}{a^2 - u^2} du$$

$$34 \int \frac{1}{u(a + bu)} du$$

$$35 \int \frac{1}{u^2(a + bu)} du$$

$$36 \int \frac{1}{u(a + bu)^2} du$$

37 If $f(x) = x/(x^2 - 2x - 3)$, find the area of the region under the graph of f from $x = 0$ to $x = 2$.

38 The region bounded by the graphs of $y = 0$, $x = 2$, $x = 3$, and $y = 1/(x-1)(4-x)$ is revolved about the y -axis. Find the volume of the resulting solid.

39 If the region described in Exercise 38 is revolved about the x -axis, find the volume of the resulting solid.

40 Suppose $g(x) = (x-c_1)(x-c_2) \cdots (x-c_n)$ for a positive integer n and distinct real numbers $c_1, c_2, \dots, c_n$. If $f(x)$ is a polynomial of degree less than n , show that

$$\frac{f(x)}{g(x)} = \frac{A_1}{x-c_1} + \frac{A_2}{x-c_2} + \cdots + \frac{A_n}{x-c_n}$$

with $A_k = f(c_k)/g'(c_k)$ for $k = 1, 2, \dots, n$. (This is a method for finding the partial fraction decomposition if the denominator can be factored into distinct linear factors.)

41 Use Exercise 40 to find the partial fraction decomposition of

$$\frac{2x^4 - x^3 - 3x^2 + 5x + 7}{x^5 - 5x^3 + 4x}$$

7.5

QUADRATIC EXPRESSIONS AND MISCELLANEOUS SUBSTITUTIONS

In this section, we study some additional techniques for finding antiderivatives. We first examine integrands that involve quadratic expressions, and then we consider a variety of integrals that can be handled by substitutions.

INTEGRALS INVOLVING QUADRATIC EXPRESSIONS

Partial fraction decompositions may lead to integrands containing an irreducible quadratic expression $ax^2 + bx + c$. If $b \neq 0$, it is sometimes necessary to complete the square as follows:

$$\begin{aligned} ax^2 + bx + c &= a \left(x^2 + \frac{b}{a}x \right) + c \\ &= a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \end{aligned}$$

The substitution $u = x + b/(2a)$ may then lead to an integrable form.

EXAMPLE 1 Evaluate $\int \frac{2x-1}{x^2-6x+13} dx$.

SOLUTION Note that the quadratic expression $x^2 - 6x + 13$ is irreducible, since $b^2 - 4ac = -16 < 0$. We complete the square as follows:

$$\begin{aligned}x^2 - 6x + 13 &= (x^2 - 6x \quad \quad) + 13 \\&= (x^2 - 6x + 9) + 13 - 9 = (x - 3)^2 + 4\end{aligned}$$

Thus,

$$\int \frac{2x-1}{x^2-6x+13} dx = \int \frac{2x-1}{(x-3)^2+4} dx.$$

We now make the substitution

$$u = x - 3, \quad x = u + 3, \quad dx = du.$$

Thus,

$$\begin{aligned}\int \frac{2x-1}{x^2-6x+13} dx &= \int \frac{2(u+3)-1}{u^2+4} du \\&= \int \frac{2u+5}{u^2+4} du \\&= \int \frac{2u}{u^2+4} du + 5 \int \frac{1}{u^2+4} du \\&= \ln(u^2+4) + \frac{5}{2} \tan^{-1} \frac{u}{2} + C \\&= \ln(x^2-6x+13) + \frac{5}{2} \tan^{-1} \frac{x-3}{2} + C.\end{aligned}$$

We may also use the technique of completing the square if a quadratic expression appears under a radical sign. In the next example, we make a trigonometric substitution after completing the square.

EXAMPLE 2 Evaluate $\int \frac{1}{\sqrt{x^2+8x+25}} dx$.

SOLUTION We complete the square for the quadratic expression as follows:

$$\begin{aligned}x^2 + 8x + 25 &= (x^2 + 8x \quad \quad) + 25 \\&= (x^2 + 8x + 16) + 25 - 16 \\&= (x + 4)^2 + 9\end{aligned}$$

Thus,
$$\int \frac{1}{\sqrt{x^2+8x+25}} dx = \int \frac{1}{\sqrt{(x+4)^2+9}} dx.$$

If we make the trigonometric substitution

$$x + 4 = 3 \tan \theta, \quad dx = 3 \sec^2 \theta d\theta,$$

then

$$\sqrt{(x+4)^2 + 9} = \sqrt{9 \tan^2 \theta + 9} = 3\sqrt{\tan^2 \theta + 1} = 3 \sec \theta$$

$$\begin{aligned} \text{and} \quad \int \frac{1}{\sqrt{x^2 + 8x + 25}} dx &= \int \frac{1}{3 \sec \theta} 3 \sec^2 \theta d\theta \\ &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C. \end{aligned}$$

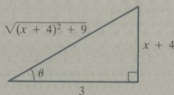
To return to the variable x , we use the triangle in Figure 7.7, obtaining

$$\begin{aligned} \int \frac{1}{\sqrt{x^2 + 8x + 25}} dx &= \ln \left| \frac{\sqrt{x^2 + 8x + 25}}{3} + \frac{x+4}{3} \right| + C \\ &= \ln |\sqrt{x^2 + 8x + 25} + x + 4| - \ln |3| + C \\ &= \ln |\sqrt{x^2 + 8x + 25} + x + 4| + K \end{aligned}$$

with $K = C - \ln 3$.

Figure 7.7

$$\tan \theta = \frac{x+4}{3}$$



MISCELLANEOUS SUBSTITUTIONS

We now consider substitutions that are useful for evaluating certain types of integrals. The first example illustrates that if an integral contains an expression of the form $\sqrt[n]{f(x)}$, then one of the substitutions $u = \sqrt[n]{f(x)}$ or $u = f(x)$ may simplify the evaluation.

EXAMPLE 3 Evaluate $\int \frac{x^3}{\sqrt[3]{x^2+4}} dx$.

SOLUTION I The substitution $u = \sqrt[3]{x^2+4}$ leads to the following equivalent equations:

$$u = \sqrt[3]{x^2+4}, \quad u^3 = x^2+4, \quad x^2 = u^3-4$$

Taking the differential of each side of the last equation, we obtain

$$2x dx = 3u^2 du, \quad \text{or} \quad x dx = \frac{3}{2} u^2 du.$$

We now substitute as follows:

$$\begin{aligned} \int \frac{x^3}{\sqrt[3]{x^2+4}} dx &= \int \frac{x^2}{\sqrt[3]{x^2+4}} \cdot x dx \\ &= \int \frac{u^3-4}{u} \cdot \frac{3}{2} u^2 du = \frac{3}{2} \int (u^4-4u) du \\ &= \frac{3}{2} \left(\frac{1}{5} u^5 - 2u^2 \right) + C = \frac{3}{10} u^2 (u^3-10) + C \\ &= \frac{3}{10} (x^2+4)^{2/3} (x^2-6) + C \end{aligned}$$

SOLUTION 2 If we substitute u for the expression *underneath* the radical, then

$$u = x^2 + 4, \quad \text{or} \quad x^2 = u - 4$$

$$\text{and} \quad 2x \, dx = du, \quad \text{or} \quad x \, dx = \frac{1}{2} du.$$

In this case, we may write

$$\begin{aligned} \int \frac{x^3}{\sqrt[3]{x^2+4}} dx &= \int \frac{x^2}{\sqrt[3]{x^2+4}} \cdot x \, dx \\ &= \int \frac{u-4}{u^{1/3}} \cdot \frac{1}{2} du = \frac{1}{2} \int (u^{2/3} - 4u^{-1/3}) du \\ &= \frac{1}{2} \left(\frac{3}{5} u^{5/3} - 6u^{2/3} \right) + C = \frac{3}{10} u^{2/3} (u - 10) + C \\ &= \frac{3}{10} (x^2 + 4)^{2/3} (x^2 - 6) + C. \end{aligned}$$

EXAMPLE 4 Evaluate $\int \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx$.

SOLUTION To obtain a substitution that will eliminate the two radicals $\sqrt{x} = x^{1/2}$ and $\sqrt[3]{x} = x^{1/3}$, we use $u = x^{1/n}$, where n is the least common denominator of $\frac{1}{2}$ and $\frac{1}{3}$. Thus, we let

$$u = x^{1/6}, \quad \text{or, equivalently,} \quad x = u^6.$$

Hence,

$$dx = 6u^5 du, \quad x^{1/2} = (u^6)^{1/2} = u^3, \quad x^{1/3} = (u^6)^{1/3} = u^2$$

and, therefore,

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx = \int \frac{1}{u^3 + u^2} 6u^5 du = 6 \int \frac{u^3}{u+1} du.$$

By long division,

$$\frac{u^3}{u+1} = u^2 - u + 1 - \frac{1}{u+1}.$$

Consequently,

$$\begin{aligned} \int \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx &= 6 \int \left(u^2 - u + 1 - \frac{1}{u+1} \right) du \\ &= 6 \left(\frac{1}{3} u^3 - \frac{1}{2} u^2 + u - \ln |u+1| \right) + C \\ &= 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6 \ln(\sqrt[6]{x} + 1) + C. \end{aligned}$$

If an integrand is a rational expression in $\sin x$ and $\cos x$, then the substitution

$$u = \tan \frac{x}{2} \quad \text{for} \quad -\pi < x < \pi$$

will transform the integrand into a rational (algebraic) expression in u . To prove this, first note that

$$\begin{aligned}\cos \frac{x}{2} &= \frac{1}{\sec(x/2)} = \frac{1}{\sqrt{1 + \tan^2(x/2)}} = \frac{1}{\sqrt{1 + u^2}} \\ \sin \frac{x}{2} &= \tan \frac{x}{2} \cos \frac{x}{2} = u \frac{1}{\sqrt{1 + u^2}}.\end{aligned}$$

Consequently,

$$\begin{aligned}\sin x &= 2 \sin \frac{x}{2} \cos \frac{x}{2} = \frac{2u}{1 + u^2} \\ \cos x &= 1 - 2 \sin^2 \frac{x}{2} = 1 - \frac{2u^2}{1 + u^2} = \frac{1 - u^2}{1 + u^2}.\end{aligned}$$

Moreover, since $x/2 = \tan^{-1} u$, we have $x = 2 \tan^{-1} u$, and, therefore,

$$dx = \frac{2}{1 + u^2} du.$$

The following theorem summarizes this discussion.

Theorem 7.6

If an integrand is a rational expression in $\sin x$ and $\cos x$, the following substitutions will produce a rational expression in u :

$$\sin x = \frac{2u}{1 + u^2}, \quad \cos x = \frac{1 - u^2}{1 + u^2}, \quad dx = \frac{2}{1 + u^2} du,$$

where $u = \tan \frac{x}{2}$ for $-\pi < x < \pi$.

EXAMPLE 5 Evaluate $\int \frac{1}{4 \sin x - 3 \cos x} dx$.

SOLUTION Applying Theorem (7.6) and simplifying the integrand yields

$$\begin{aligned}\int \frac{1}{4 \sin x - 3 \cos x} dx &= \int \frac{1}{4 \left(\frac{2u}{1 + u^2} \right) - 3 \left(\frac{1 - u^2}{1 + u^2} \right)} \cdot \frac{2}{1 + u^2} du \\ &= \int \frac{2}{8u - 3(1 - u^2)} du \\ &= 2 \int \frac{1}{3u^2 + 8u - 3} du.\end{aligned}$$

Using partial fractions, we have

$$\frac{1}{3u^2 + 8u - 3} = \frac{1}{10} \left(\frac{3}{3u - 1} - \frac{1}{u + 3} \right)$$

and hence

$$\begin{aligned}\int \frac{1}{4 \sin x - 3 \cos x} dx &= \frac{1}{5} \int \left(\frac{3}{3u-1} - \frac{1}{u+3} \right) du \\&= \frac{1}{5} (\ln |3u-1| - \ln |u+3|) + C \\&= \frac{1}{5} \ln \left| \frac{3u-1}{u+3} \right| + C \\&= \frac{1}{5} \ln \left| \frac{3 \tan(x/2) - 1}{\tan(x/2) + 3} \right| + C.\end{aligned}$$

Theorem (7.6) may be used for any integrand that is a rational expression in $\sin x$ and $\cos x$. However, it is also important to consider simpler substitutions, as illustrated in the next example.

EXAMPLE 6 Evaluate $\int \frac{\cos x}{1 + \sin^2 x} dx$.

SOLUTION We could use the formulas in Theorem (7.6) to change the integrand into a rational expression in u . The following substitution is simpler:

$$u = \sin x, \quad du = \cos x \, dx$$

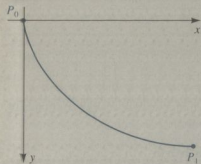
Thus,

$$\begin{aligned}\int \frac{\cos x}{1 + \sin^2 x} dx &= \int \frac{1}{1 + u^2} du \\&= \arctan u + C \\&= \arctan \sin x + C.\end{aligned}$$

The evaluation of an integral may well involve the application of several techniques in succession. John Bernoulli's solution of the *brachistochrone problem*, for example, requires an algebraic substitution, a trigonometric substitution, and the use of several trigonometric identities. In 1696, Bernoulli published this problem as a challenge to other mathematicians: Find among all smooth curves lying in a vertical plane and connecting a given higher point P_0 to a given lower point P_1 , the curve along which a particle will slide in the shortest possible time.

In Figure 7.8, we have set up a coordinate system with the higher point P_0 at the origin and with the positive direction of the y -axis downward. Figure 7.8 also shows a curve joining the points. Such a curve occurs in planning the design of a ski jump, for example. Bernoulli wished to find explicitly an expression for the function $y = f(x)$ whose graph is the curve of fastest descent from P_0 to P_1 . Assuming that gravity is the only force acting on the particle, Bernoulli used ideas from optics, mechanics,

Figure 7.8



and calculus to discover that the function f must satisfy the differential equation

$$y \left[1 + \left(\frac{dy}{dx} \right)^2 \right] = c$$

for some constant c .

We may rewrite this equation as

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right] = \frac{c}{y}$$

or, equivalently,

$$\left(\frac{dy}{dx} \right)^2 = \frac{c}{y} - 1 = \frac{c - y}{y},$$

so that

$$\frac{dy}{dx} = \sqrt{\frac{c - y}{y}},$$

where we have chosen the positive square root, since in our coordinate system y increases as x increases, making $dy/dx > 0$. Since the right side of this differential equation is a function only of the variable y , we separate the variables and obtain

$$\int dx = \int \sqrt{\frac{y}{c - y}} dy,$$

so that

$$x = \int \sqrt{\frac{y}{c - y}} dy.$$

To carry out the integration of the right side, we first make the algebraic substitution $u = \sqrt{y/(c - y)}$, so that $u^2 = y/(c - y)$, which gives $y = cu^2/(1 + u^2)$ and, with differentiation, $dy = [2cu/(1 + u^2)^2] du$. Thus,

$$x = \int \sqrt{\frac{y}{c - y}} dy = \int u \cdot \frac{2cu}{(1 + u^2)^2} du = \int \frac{2cu^2}{(1 + u^2)^2} du.$$

To evaluate the last integral, we make the trigonometric substitution $u = \tan \phi$, $du = \sec^2 \phi d\phi$. Thus, the integrand becomes

$$\frac{2cu^2}{(1 + u^2)^2} = \frac{2c \tan^2 \phi}{(1 + \tan^2 \phi)^2} = \frac{2c \tan^2 \phi}{(\sec^2 \phi)^2}$$

so that

$$x = \int \frac{2cu^2}{(1 + u^2)^2} du = \int \frac{2c \tan^2 \phi}{(\sec^2 \phi)^2} \sec^2 \phi d\phi = \int 2c \sin^2 \phi d\phi.$$

Our final substitution is to use a variation of the double-angle formula for the cosine, $2 \sin^2 \phi = 1 - \cos 2\phi$, obtaining

$$x = c \int (1 - \cos 2\phi) d\phi.$$

Integration yields

$$x = \frac{c}{2}(2\phi - \sin 2\phi) + D$$

for some constant D . Substituting $\phi = 0$ into this equation, we see that D is equal to the value of x when $\phi = 0$. At $\phi = 0$, $u = \tan 0 = 0$ and hence $y = cu^2/(1+u^2) = 0$. But when $y = 0$, we also have $x = 0$ because the point $P_0(0, 0)$ lies on the curve. Thus, $D = 0$.

Using $u = \tan \phi$ gives us a formula for y :

$$y = \frac{cu^2}{1+u^2} = \frac{c \tan^2 \phi}{1 + \tan^2 \phi} = \frac{c \tan^2 \phi}{\sec^2 \phi} = c \sin^2 \phi = \frac{c}{2}(1 - \cos 2\phi)$$

As a final simplification, we let $a = c/2$ and $\theta = 2\phi$. The curve of fastest descent is then described by the set of points $P(x, y)$, where x and y are given by a pair of equations:

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta)$$

Here we have described the curve by giving separate equations for the x - and y -coordinates in terms of a third variable, θ . This description is called a *parametric representation* of a curve; we will study such representations in more detail in Chapter 9. In this case, we may regard θ as representing time. At $\theta = 0$, $x = 0$ and $y = 0$, and the particle is at the point P_0 . As θ increases, the particle moves down the curve toward the point P_1 .

EXERCISES 7.5

Exer. 1–18: Evaluate the integral.

- 1 $\int \frac{1}{(x+1)^2+4} dx$
- 2 $\int \frac{1}{\sqrt{16-(x-3)^2}} dx$
- 3 $\int \frac{1}{x^2-4x+8} dx$
- 4 $\int \frac{1}{x^2-2x+2} dx$
- 5 $\int \frac{1}{\sqrt{4x-x^2}} dx$
- 6 $\int \frac{1}{\sqrt{7+6x-x^2}} dx$
- 7 $\int \frac{2x+3}{\sqrt{9-8x-x^2}} dx$
- 8 $\int \frac{x+5}{9x^2+6x+17} dx$
- 9 $\int \frac{1}{(x^2+4x+5)^2} dx$
- 10 $\int \frac{1}{(x^2-6x+34)^{3/2}} dx$
- 11 $\int \frac{1}{(x^2+6x+13)^{3/2}} dx$
- 12 $\int \sqrt{x(6-x)} dx$
- 13 $\int \frac{1}{2x^2-3x+9} dx$
- 14 $\int \frac{2x}{(x^2+2x+5)^2} dx$
- 15 $\int \frac{e^x}{e^{2x}+3e^x+2} dx$
- 16 $\int \sqrt{x^2+10x} dx$

$$17 \int_2^3 \frac{x^2-4x+6}{x^2-4x+5} dx \quad 18 \int_0^1 \frac{x-1}{x^2+x+1} dx$$

19 Find the area of the region bounded by the graphs of $y = 1/(x^2+4x+29)$, $y = 0$, $x = -2$, and $x = 3$.

20 The region bounded by the graph of

$$y = 1/(x^2+2x+10),$$

the coordinate axes, and the line $x = 2$ is revolved about the x -axis. Find the volume of the resulting solid.

Exer. 21–46: Evaluate the integral.

- 21 $\int x\sqrt{x+9} dx$
- 22 $\int x^2\sqrt{2x+1} dx$
- 23 $\int \frac{x}{\sqrt[3]{3x+2}} dx$
- 24 $\int \frac{5x}{(x+3)^{2/3}} dx$
- 25 $\int_4^9 \frac{1}{\sqrt{x}+4} dx$
- 26 $\int_0^{25} \frac{1}{\sqrt{4+\sqrt{x}}} dx$

$$27 \int \frac{\sqrt{x}}{1+\sqrt[3]{x}} dx$$

$$29 \int \frac{1}{(x+1)\sqrt{x-2}} dx$$

$$31 \int \frac{x+1}{(x+4)^{1/3}} dx$$

$$33 \int e^{3x} \sqrt{1+e^x} dx$$

$$35 \int \frac{e^{2x}}{e^x+4} dx$$

$$37 \int \sin \sqrt{x+4} dx$$

$$39 \int_2^3 \frac{x}{(x-1)^6} dx$$

$$41 \int \frac{\sin x}{\cos x(\cos x-1)} dx$$

$$43 \int \frac{e^x}{e^{2x}-1} dx$$

$$28 \int \frac{1}{\sqrt[4]{x}+\sqrt[3]{x}} dx$$

$$30 \int_0^4 \frac{2x+3}{\sqrt{1+2x}} dx$$

$$32 \int \frac{x^{1/3}+1}{x^{1/3}-1} dx$$

$$34 \int \frac{e^{2x}}{\sqrt[3]{1+e^x}} dx$$

$$36 \int \frac{\sin 2x}{\sqrt{1+\sin x}} dx$$

$$38 \int \sqrt{x} e^{\sqrt{x}} dx$$

$$40 \int \frac{x^2}{(3x+4)^{10}} dx$$

$$42 \int \frac{\cos x}{\sin^2 x - \sin x - 2} dx$$

$$44 \int \frac{1}{e^x + e^{-x}} dx$$

$$45 \int \frac{\sin 2x}{\sin^2 x - 2 \sin x - 8} dx$$

$$46 \int \frac{\sin x}{5 \cos x + \cos^2 x} dx$$

Exer. 47–52: Use Theorem (7.6) to evaluate the integral.

$$47 \int \frac{1}{2+\sin x} dx$$

$$48 \int \frac{1}{3+2\cos x} dx$$

$$49 \int \frac{1}{1+\sin x + \cos x} dx$$

$$50 \int \frac{1}{\tan x + \sin x} dx$$

$$51 \int \frac{\sec x}{4-3\tan x} dx$$

$$52 \int \frac{1}{\sin x - \sqrt{3} \cos x} dx$$

Exer. 53–54: Use Theorem (7.6) to derive the formula.

$$53 \int \sec x dx = \ln \left| \frac{1 + \tan \frac{1}{2}x}{1 - \tan \frac{1}{2}x} \right| + C$$

$$54 \int \csc x dx = \frac{1}{2} \ln \left(\frac{1 - \cos x}{1 + \cos x} \right) + C$$

7.6

TABLES OF INTEGRALS AND COMPUTER ALGEBRA SYSTEMS